

# Maria Dodgson

Biochemistry, BSc. Honours

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## SUMMARY

Equipped with an education in Molecular Biology and Green Chemistry, as well as experience in research, project coordination and scientific publication, I am eager to contribute these skills in a meaningful position. With excellent communication skills and life science laboratory capabilities, I am well adapted to working as an energetic member of a team as well as a proficient self-starter.

## WORK EXPERIENCE

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### Laboratory Analyst

*ALS Canada Ltd.*

**May 2025 – Present**

*Burnaby, BC*

- Performed high-throughput trace metals analysis of 300+ samples daily using ICP-OES and ICP-MS in compliance with SOPs and client-specific protocols across environmental, pharmaceutical, and agricultural sectors.
- Conducted routine maintenance, troubleshooting, and method optimization of ICP-OES and ICP-MS instruments using Microsoft Excel, Agilent software, and Angle for data analysis and system calibration.
- Prepared precise sample dilutions (up to 100,000x), sequenced runs, and performed quality control checks on 500+ samples per day, maintaining strict adherence to QA/QC guidelines and regulatory standards.

### Client Care Coordinator

*Simply Radiant*

**October 2023 – May 2025**

*Mount Pearl, NL*

- Managed patient forms, appointments, and provided personalized care to enhance client satisfaction and retention for 500+ clients in a medical aesthetics environment.
- Processed payments, maintained accurate patient records, and implemented an improved method to record inventory, ensuring efficient clinical operations.
- Designed and executed social media content, including a brand logo, promotions, testimonials, and treatment information, increasing clinic visibility and engagement across Instagram and TikTok platforms.

### Research Assistant

*Memorial University of Newfoundland*

**May 2022 – July 2023**

*St. John's, NL*

- Performed tissue culture and transfection of human embryonic kidney cells to study the effect of RNA mutations on the development of leukemia.
- Prepared protein, DNA, and RNA extractions for quantification with SDS-PAGE, Western Blot, and quantitative PCR to determine gene copy number and gene expression levels.
- Generated recombinant DNA plasmids using restriction digestion, electrophoresis, and transformation to increase RNA expression and determine biomarkers of cancer.
- Conducted experimental research in collaboration with Dr. Kakumani for a publication in *Nucleic Acids Research*, focusing on the synthesis of a specific miRNA in leukemia.

### Research Assistant

*Memorial University of Newfoundland*

**Sept. 2021 – April 2022**

*St. John's, NL*

- Analyzed the absorbance of toxic materials in calcium carbonate samples for solutions to pollutant removal in water treatment.
- Characterized seaweed polysaccharides with UV-Vis spectroscopy for commercial antioxidant dermatology products.
- Conducted literature research on mussel shells as a renewable source of calcium carbonate and on

green solvent exfoliation of biochars to gain expertise in sustainable environment solutions.

## EDUCATION

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### BSc. Honours in Biochemistry, Minor in Chemistry

Feb. 2024

*Memorial University of Newfoundland*

*St. John's, NL*

- **Thesis: Regulatory role of CSDE1 in the expression of miR-708**
- **Supervisor: Dr. Pavan Kakumani**
- Conducted extensive research projects in cancer and molecular biology, collaborating with Dr. Kakumani to produce findings reports and results for a scientific publication.
- Relevant course work completed in Molecular Biology, Research Techniques, Immunology, and Medicine.

Developed strong skills in research, scientific writing, and data analysis, with a focus on optimizing efficient and effective solutions to complex challenges in disease, pharmaceuticals, and environmental sustainability.

## PUBLICATIONS

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### CSDE1 promotes miR-451 biogenesis

Sept. 2023

Kakumani, P. K., Ko, Y., Ramakrishna, S., Christopher, G., **Dodgson, M.**, Shrinet, J., Harvey, L. M., Shin, C., & Simard, M. J. (2023). CSDE1 promotes miR-451 biogenesis. *Nucleic acids research*, 51(17), 9385–9396. <https://doi.org/10.1093/nar/gkad619>

## VOLUNTEER EXPERIENCE

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### Social Media Executive

May 2023

*Memorial University Save the Sea Society*

*St. John's, NL*

- Managed event planning and social media strategy to advance environmental advocacy efforts, engaging communities and raising awareness.

### Forensic Science Education Program

May 2023

*Let's Talk Science*

*St. John's, NL*

- Mentored students in forensic and anthropological DNA analysis techniques, fostering skills in investigative methodologies.
- Conducted forensic analysis using PCR assays to identify physical characteristics and establish profiles from unknown biological samples.

### Scientific Endeavors in Academia Conference

April 2022

*Memorial University of Newfoundland*

*St. John's, NL*

- Managed participant registration and provided comprehensive support for keynote speakers, ensuring smooth logistics and a seamless event experience.

### Global Women's Breakfast

Feb. 2022

*International Union of Pure and Applied Chemistry (IUPAC)*

*St. John's, NL*

- Led community outreach initiatives and coordinated volunteer efforts to foster engagement and event support aimed at empowering women to pursue careers in science and STEM fields.

## SKILLS

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- **Technical:** ReactJS; CSS3; HTML5; JavaScript; GitCLI; GitHub; User Interface Design

- **Software:** Microsoft Office Suite; Agilent, Angle, GraphPad Prism; Coot; ImageJ; BLAST; NEBaseChanger; RNAfold Webserver; PyMOL
- **Experimental:** Data analysis; ICP-MS, ICP-OES, Cell and Bacterial Culture; qPCR; Western Blot; DNA/RNA Isolation; Spectroscopy; Electrophoresis; Bradford Assay
- **Languages:** Intermediate French
- **Software:** React, JavaScript, TypeScript, CSS3, HTML5, Github, Microsoft Office Suite; GraphPad Prism; Coot; ImageJ; BLAST; NEBaseChanger; RNAfold Webserver; PyMOL

**REFERENCES AVAILABLE UPON REQUEST.**